Model Risk Factory

The Agent does the reading, porting and number-crunching of a model review. Reviewers keep every decision, and every number traces to evidence.
Challenge 04 · EXL × UCD Data & AI Hackathon 2026 · competition prototype

- Document: three legacy R models inventoried from code and stale notes. Conflicts and Unknowns are kept, each with path, line and SHA-256.
- Migrate: MDL_CREDIT_V1 ported to Python by a real gateway Agent. Max |Δp| 2.220e-16 against R on 12,000 records (HIST-UCI).
- Validate: performance, calibration, drift and fairness computed by Python. 191 report values each resolve to a JSON path (HIST-UCI).
- Control: three hash-bound review gates. Any change to code, config or candidate invalidates earlier approvals.

results come only from runs recorded in the repository. HIST-UCI = v2-20261109T210903Z-3dcfab73 (UCI, seed 46, historical); SYN-SEED47 = dev_runs/v2-20261110T182506Z-9acb1487 (synthetic). Release state: slide 9.

The client problem

CLIENT-CONTEXT (from the challenge brief, not measured here): a bank runs ~250 models; documentation drifts away from code; a model review takes 8–12 weeks; PRA SS1/23 sets the expectation that models are documented, validated, monitored and independently reviewed.

What that costs in practice, and what this prototype addresses:

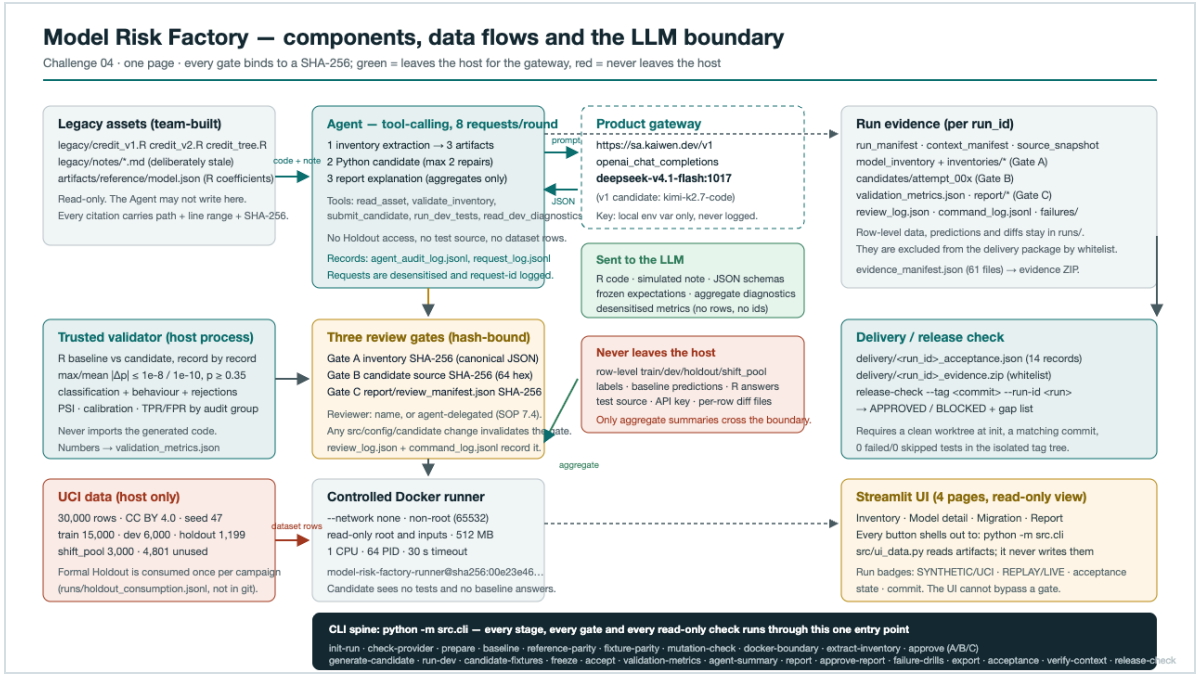
- The document and the code disagree, and nobody can say which one is right without reading both.
- Re-implementation is checked by eye, sampled, or "the AUC looks fine".
- Validation evidence is rebuilt per review instead of being produced by the pipeline.
- Drift is found by a spreadsheet refresh, not by a metric with a recorded threshold.

docs/01_比赛背景与官方要求.md and the official PDF. The 250-model / 8–12-week figures are client backdrop only and are never presented as this project's results.

Method and architecture

The Agent proposes artifacts, deterministic checks validate them, and reviewers record decisions at three gates.

- Document → Agent returns a schema-validated inventory; Gate A approves its hash. Redevelop → Agent writes the Python candidate; Gate B approves the 64-hex source hash before anything runs.
- Validate → candidate runs in a no-network, read-only container; a trusted validator compares it with the R baseline. Monitor → drift, calibration and fairness; Gate C approves the rendered report.
- Hard boundary: Holdout inputs and answers never enter the Agent context; only aggregate, desensitised metrics do.

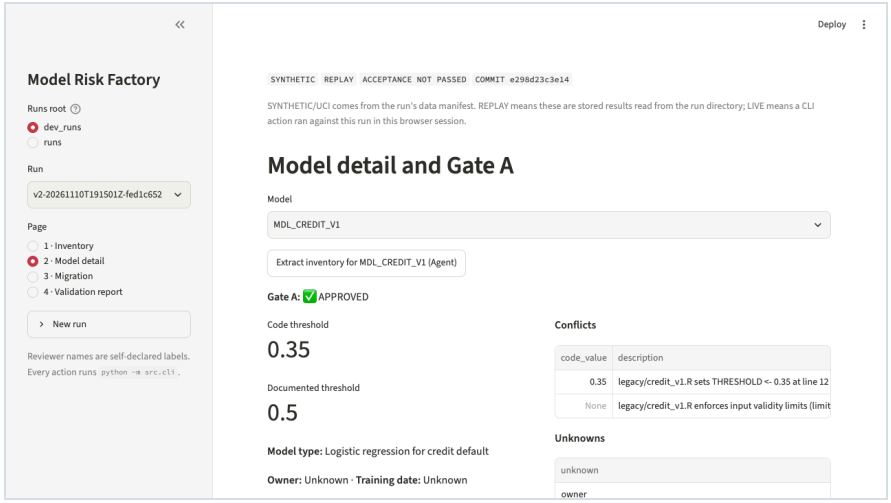


architecture detail in materials/architecture.md. Gate definitions: docs/02_项目SOP.md §7.4.

Inventory and documented gaps

Three models documented from their real code and their real (deliberately stale) notes.

- MDL_CREDIT_V1: code threshold 0.35 vs documented 0.50 — kept as a conflict, never auto-normalised; 14 verified sources.
- MDL_CREDIT_V2 and MDL_CREDIT_TREE: undocumented input limits, transformations and hyperparameters recorded as conflicts; owner and training date stay Unknown.
- Each confirmed conflict carries file path, line, excerpt and SHA-256. Equal code and document thresholds cannot be recorded as a conflict.
- HIST-UCI: 3 inventories, 3 Gate A approvals, 4 Unknown fields each.



HIST-UCI model_inventory.json, inventories/.json, inventory_validation.json; rule set src/inventory.py + config/inventory_expectations.json; regression tests/test_v3_gap_inventory_conflicts.py.*

R to Python parity

The candidate reuses the exported R coefficients, medians, feature order and the 0.35 threshold. Nothing is retrained.

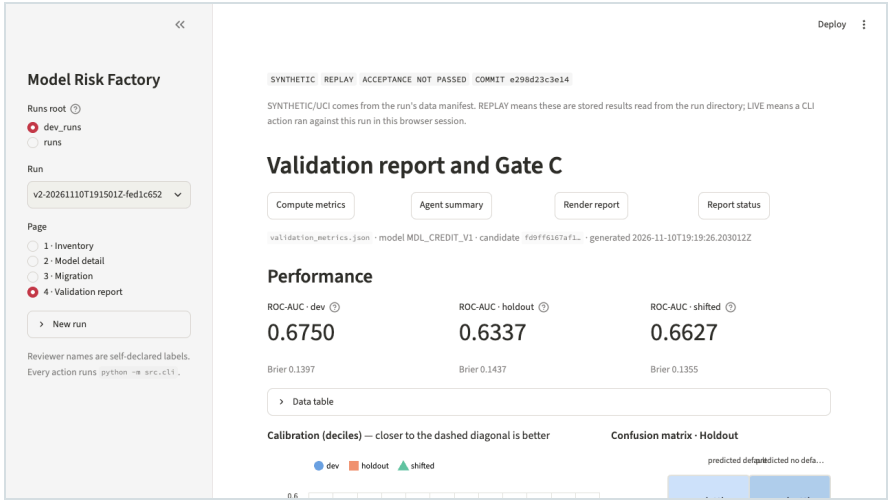
- HIST-UCI: 6,000 development records and 6,000 Holdout records compared one by one — max |Δp| 2.220e-16, mean |Δp| 2.707e-17, 0 classification mismatches, 0 behaviour mismatches (tolerance 1e-8 / 1e-10).
- Behaviour is tested too: missing values, imputation flags, invalid rows and rejection codes match record by record, plus row-order, column-order and structural-refusal fixtures.
- A deliberately altered coefficient (+0.001 on delay_1) is caught by the fixed validator; the original baseline and test hashes stay unchanged.
- The Agent never sees the Holdout: it only gets development parity summaries and aggregate diagnostics.

HIST-UCI parity//parity_summary.json, mutation/mutation_summary.json, dev/attempt_000/fixtures/; thresholds config/model_config.json.*

Validation and monitoring

Python computes the metrics and renders the conclusions. The Agent selects a review focus and its structured facts are checked against the metrics.

- Performance (HIST-UCI Holdout, 6,000 rows): ROC-AUC 0.7529, Brier 0.1422, decile calibration table, confusion matrix at p ≥ 0.35 (TP 621 / FP 501 / FN 701 / TN 4177).
- Stability: weighted drift batch vs dev reference — prediction PSI 0.0170, worst feature PSI 0.0799 (util_1); attention 0.10 / review 0.25 are project demo rules, not a regulatory standard.
- Fairness audit: age_band 21–25 TPR 0.5477 vs reference 0.4392 (+0.1086), sex=1 TPR difference –0.0387; small groups are marked LOW_SAMPLE and rates are not assessable.
- Limits sit next to the numbers: same-period hold-out (not a back-test), drift is a sensitivity analysis, no cross-time back-test. Gate C re-renders and compares every cell (SYN-SEED47: 196 values, 0 unresolved).



HIST-UCI validation_metrics.json, report/validation_report.md; SYN-SEED47 report/values_index.csv, report/review_manifest.json.

What the reviewer no longer does by hand

Each line is a task a validator normally does by hand, with the measured automated time (SYN-SEED47, gateway deepseek-v4.1-flash:1017).

- Read code and notes, write the inventory and list the conflicts: 12.7 s / 31.6 s / 40.3 s per model. A deterministic rule rejected one incomplete answer automatically.
- Re-implement the model in Python: 66.1 s for the candidate. Prove it matches R record by record: run by the pipeline, with no sampling or checking by eye.
- Recompute every report number and check every table cell: done at Gate C. Draft the review summary: 9.8 s.
- End to end: 29 commands and 3.49 min of automated work for 3 inventories, a migration, validation, report and export. Reviewer time is spent on the decisions at the gates.
- Not claimed yet: a speed-up percentage. The paired manual-vs-Agent study (protocol in materials/impact_measurements_README.md) has 0 completed pairs.

SYN-SEED47 command_log.jsonl timings, agent_audit_log.jsonl, provider/request_log.jsonl; materials/impact_measurements.csv + its README.

Responsible AI

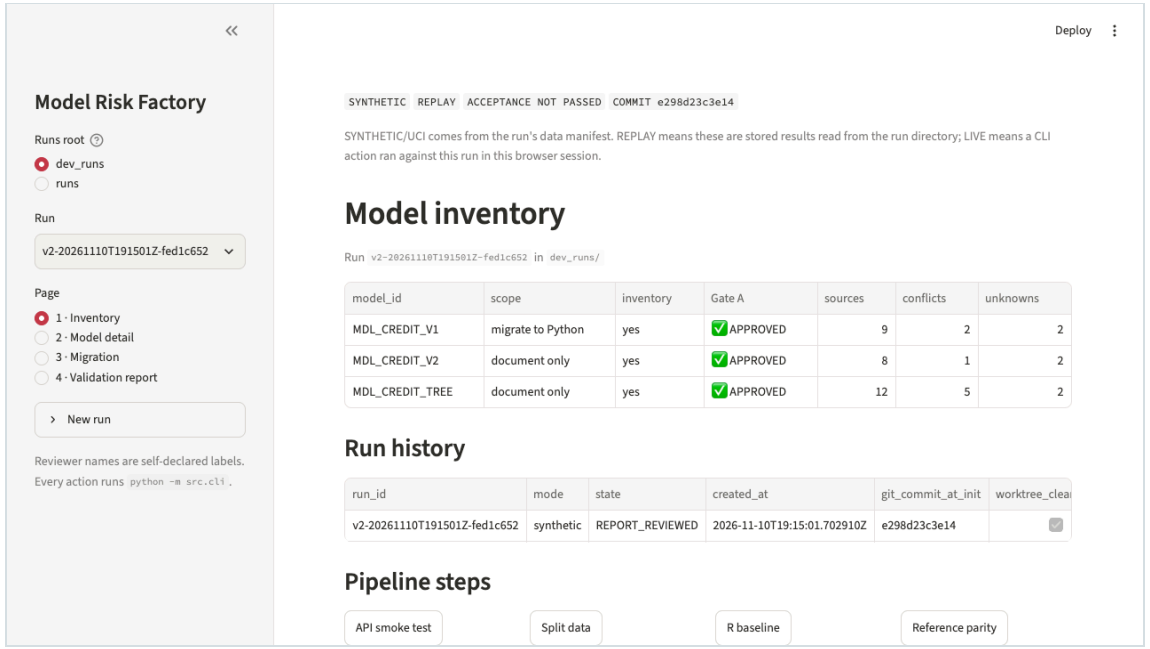
- Data: UCI Default of Credit Card Clients (CC BY 4.0), a 2005 Taiwan sample; no direct identifiers but demographic attributes exist. Competition use permission is still recorded as unconfirmed (uci_competition_use_approved=false).
- Privacy: demographics are audit-only and never model inputs; no raw rows, labels or Holdout content are sent to the LLM — only R code, the simulated note, schemas and aggregate diagnostics; API keys live in local environment variables only.
- Bias: TPR/FPR by age_band and sex are audit signals, not a fairness verdict; small groups are marked not assessable; the model is not claimed to be fair.
- Limitations: not production-ready, not independently validated, not a regulatory certification; no cross-time back-test; report conclusions come from computed metrics and are checked again at Gate C.
- Safeguards: generated code runs only in a no-network, read-only, non-root container with memory/CPU/PID/time limits; gates A/B/C bind to hashes and invalidate on any source, config or candidate change.

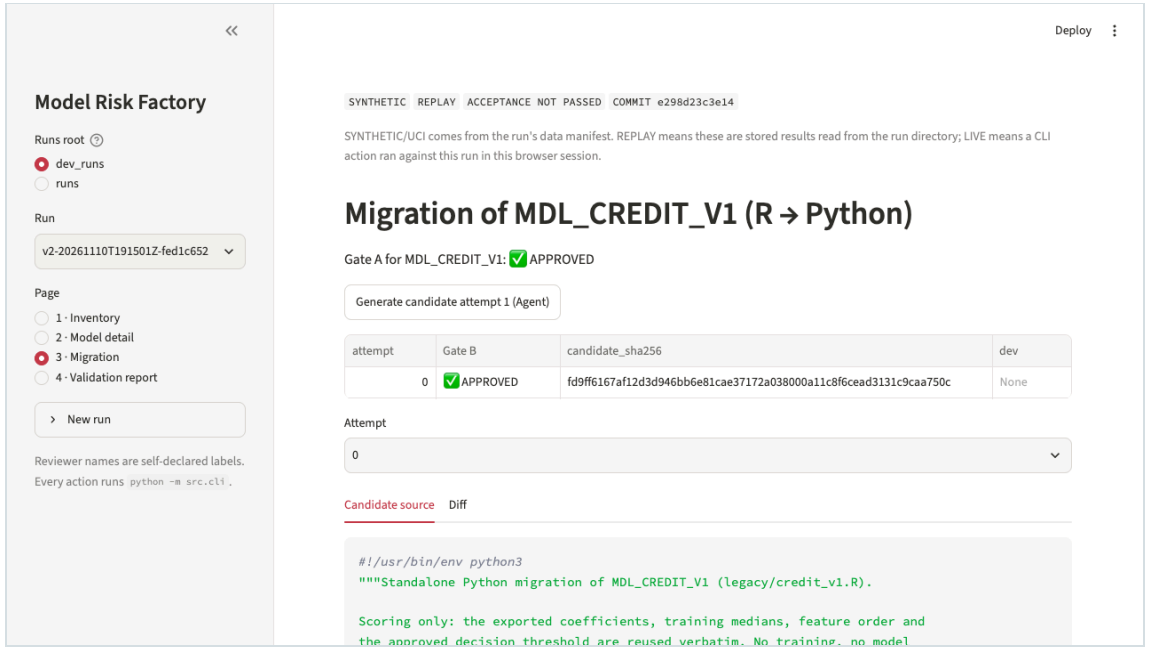
HIST-UCI boundary/boundary_report.json, review_log.json, report/validation_report.md; materials/responsible_ai_note.md.

Prototype walkthrough and release state

Four Streamlit pages over the same CLI the pipeline uses — every button runs python -m src.cli, so the UI cannot bypass a gate.

- Inventory, Model detail (Gate A), Migration (Gate B + Docker), Validation report (Gate C + downloads).
- The 1:51 demo replays a synthetic run. The current drill pauses at every gate for a reviewer decision.
- Release state, stated plainly: v1 accepted locally (tag v1 → 70858cd, 7/7 PASS, not pushed). v2 BLOCKED, because its evidence matches no commit (17 gaps). The single v3 formal run (seed 47) is scripted and preflight-checked, and waits on the organiser's UCI permission, ledger confirmation and named reviewers.





docs/v3/00_baseline_status.md, docs/v3/03_release_readiness.md, docs/v3/02_new_environment.md; screenshots in docs/v3/ui_screens/.

Roadmap and disclosure

Why this is different: approvals bind exact hashes and expire on any change; every report number resolves to a JSON path; parity is proven record by record and mutation-tested.

- Short term: one authorised formal seed-47 run on the frozen commit, human Gate A/B/C sign-off, recorded demo of the same chain.
- Then: SAS/other formats behind the same inventory contract; batch queue across ~250 models; MLflow/Azure ML as an evidence sink, not a replacement for the gates; genuine out-of-time data for back-testing.
- AI tools used: Claude Code / Kimi Code CLI (implementation, tests, docs) and this delivery Agent; in-product gateway sa.kaiwen.dev with deepseek-v4.1-flash:1017 for inventory extraction, candidate generation and report explanation (v1 candidate used kimi-k2.7-code).
- Existing code and data: R glm/rpart/jsonlite, Python NumPy/pandas/Pydantic/pytest/Docker/Streamlit, the UCI dataset, and the three team-built R models plus their simulated notes.
- What is human-written: the trusted Python reference predictor and the fixed tests. What is Agent-written: the migration candidate, the inventories and the report narrative.

README.md (AI tool and existing-code disclosure), HIST-UCI provider/request_log.jsonl, agent_audit_log.jsonl, V1-UCI candidate.json.